CO3020 – Computer Organization

Assignment #2 – MIPS Individual Programming Assignment

Date: 2026/10/26 Deadline: 2026/11/09

Write a MIPS program in MARS to print a Fibonacci sequence using recursion. Record a short video explaining your code and showing how registers and the stack support recursive calls.

A. Program requirements

Use the following definition:

F(0) = 0
F(1) = 1
F(n) = F(n − 1) + F(n − 2), for n ≥ 2

- Read an integer n from the user. Accept 0 ≤ n ≤ 20. If n is outside this range, display an error message and ask again. You do not need to handle non-integer input.
- Implement a **recursive procedure** named fib. For n ≥ 2, call fib(n − 1) and fib(n − 2), then add their results. Do not replace this recursive calculation with a loop or a precomputed lookup table.
- Include C/C++ pseudocode that matches your MIPS program.
- Pass n in $a0 and return F(n) in $v0. Use jal for calls and jr $ra for returns.
- Save the return address and any values needed after a recursive call. Restore any $s registers you modify, and restore $sp before returning. Describe your stack frame layout in the report.
- Print F(0), F(1), …, F(n), then exit. This produces n + 1 terms. Separate the terms with commas or spaces.
- The program must assemble and run in MARS.

B. Example interaction:

Enter n (0–20): 21

Invalid input. Please enter a value from 0 to 20.

Enter n (0–20): 5

Fibonacci sequence: 0, 1, 1, 2, 3, 5

- ✓ **Test n = 0, 1, 5, and 10.**
- ✓ **Also test an integer outside the accepted range and confirm that the program asks again.**

C. Video explanation

Submit a 3–10 minute screen recording with your own spoken explanation in English. Show your code and MARS. Focus on why the code works rather than reading each instruction.

- ✧ **Assessment focuses on technical understanding and clear explanations that match your code. Video editing, English grammar, pronunciation, and fluency are not graded. Your video will be partially evaluated by your peers.**

✓ **In the video, you should:**

1. **Demonstrate your program:** Show the output for n = 0, 1, 5, and 10. Enter 50 to demonstrate input validation.
2. **Explain your recursive logic:** Use the corresponding C/C++ code to explain how your MIPS implementation handles the base cases and combines the results of F(n − 1) and F(n − 2).
3. **Demonstrate stack usage:** Use F(3) with breakpoints or single stepping in MARS to show how a stack frame is created and released. Identify the saved values and changes to $sp. Explain how the first recursive result is kept safe during the second call and how the saved $ra is used to return to the correct instruction.

D. Submission

- **Source code: StudentID_Assignment2_fib.asm**
- **Report: StudentID_Assignment2_fib.pdf, including these comments:**
  - Your name and student ID
  - C/C++ pseudocode matching your MIPS program
  - MIPS assembly source code
  - Register usage and a labeled stack frame layout
  - Test inputs and results, including an invalid input
  - A brief AI-use statement
- **Video: StudentID_Assignment2_fib.mp4, or an accessible YouTube video link.**
- Submit all three items through ee-class. Include your name, student ID, and AI-use statement in the source comments as well.

E. Assessment

Criterion	Weight
Video explanation and MARS demonstration	40%
Program correctness, including stack management and calling convention	30%
Project report	20%
Peer Evaluation	10%

F. Disclosure Use of AI tools

You may use AI tools to support learning and debugging. You are responsible for understanding and checking all submitted code. Explain the program in your own words and voices and connect your explanation to its actual execution. Briefly state which tool you used and what assistance it provided, or state that no AI tools were used.